ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0	Common difference is -1; $a_{p+q}=0$
2	**(d)** 54	$a_n=\frac{2S_n}{n}-a$
3	**(b)** 46	$a_n=S_n-S_{n-1}$
4	**(c)** 4	$d=2p$
5	**(d)** 1325	Find n first
6	**(a)** 11	$S_n=0\Rightarrow n=1-\frac{2a}{d}$
7	**(d)** 0	Common difference is -1; $a_{p+q}=0$
8	**(b)** 59	$a_n=\frac{2S_n}{n}-a$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$7-3=11-7=15-11=4$; R is the defining property of an AP.
10	**(c)** A is true but R is false.	$a_{10}=2+9\times3=29$, so A is true; the correct formula is $a+(n-1)d$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a+3d=9$, $a+8d=4$ (1); $d=-1$, $a=12$ so $a_{13}=0$ (1).	1+1
12	$a_1=S_1=7$ (1); $S_2=20$, $a_2=13$, $d=6$ (1).	1+1
13	$2a+8d=34$, $2a+9d=37$ (1); $d=3$, $a=5$ (1).	1+1
14	$a+4d=12$, $a+11d=5$ (1); $d=-1$, $a=16$ so $a_{17}=0$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{6}{2}(2a+5d)=75$ and $\frac{10}{2}(2a+9d)=185$ (1); simplify to linear equations in a,d (1); $a=5$, $d=3$, AP $5,8,11,14,\ldots$ (1).	1+1+1
16	$d=4$, $a=6$ (2); $S_{16}=576$ (1).	2+1
17	Terms form an AP with $d=7$ (1); count and endpoints (1); sum $=2916$ (1).	1+1+1
	OR: $\frac{5}{2}(2a+4d)=50$ and $\frac{9}{2}(2a+8d)=162$ (1); simplify to linear equations in a,d (1); $a=2$, $d=4$, AP $2,6,10,14,\ldots$ (1).	1+1+1
18	$d=5$, $a=9$ (2); $S_{20}=1130$ (1).	2+1
	OR: Terms form an AP with $d=9$ (1); count and endpoints (1); sum $=4867$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $a=100$, $d=20$ (1). (ii) $a_{10}=100+9\times20=₹280$ (1). (iii) $S_{12}=\frac{12}{2}[2(100)+11(20)]=6\times420=₹2520$ (1). (iv) $100+(n-1)20=400\Rightarrow n=16$ (1); $S_{16}=\frac{16}{2}(100+400)=₹4000$ (1).	1+1+1+2
20	AP with $a=90$, $l=30$, $n=31$ (1); $S_n=\frac{31}{2}(90+30)=1860$ cm (2); middle term $=\frac{90+30}{2}=60$ cm (2).	1+2+2
	OR: AP: $a=600$, $d=45$, $n=40$ (1); last $=a+(n-1)d=2355$ (2); total $=S_n=\frac{40}{2}(600+2355)=59100$ (2).	1+2+2

Mistakes to avoid:

- Using $a_n=a+nd$ instead of $a+(n-1)d$.
- Writing $S_n=\frac{n}{2}[a+(n-1)d]$; the bracket needs $2a$.
- Taking d as a_1-a_2; it is a_2-a_1 (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1116	Find n first
2	**(b)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
3	**(a)** 0	Common difference is -1; $a_{p+q} = 0$
4	**(a)** 52	$a_n = \frac{2S_n}{n} - a$
5	**(c)** 91	$a_n = S_n - S_{n-1}$
6	**(d)** 8	$d = 2p$
7	**(b)** 1065	Find n first
8	**(b)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
9	**(d)** A is false but R is true.	Differences are $3, 5, 7, \ldots$, which are not constant: A false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$S_{10} = \frac{10}{2}(1 + 10) = 55$ is true and R is a true formula, but R is not what gives the sum.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_1 = S_1 = 7$ (1); $S_2 = 18$, $a_2 = 11$, $d = 4$ (1).	1+1
12	$2a + 8d = 36$, $2a + 9d = 40$ (1); $d = 4$, $a = 2$ (1).	1+1
13	$a + 5d = 11$, $a + 10d = 6$ (1); $d = -1$, $a = 16$ so $a_{17} = 0$ (1).	1+1
14	$a_1 = S_1 = 5$ (1); $S_2 = 18$, $a_2 = 13$, $d = 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{7}{2}(2a + 6d) = 98$ and $\frac{12}{2}(2a + 11d) = 228$ (1); simplify to linear equations in a, d (1); $a = 8$, $d = 2$, AP $8, 10, 12, 14, \ldots$ (1).	1+1+1
16	$d = 3$, $a = 12$ (2); $S_{14} = 441$ (1).	2+1
17	Terms form an AP with $d = 6$ (1); count and endpoints (1); sum $= 1824$ (1).	1+1+1
	OR: $\frac{4}{2}(2a + 3d) = 42$ and $\frac{8}{2}(2a + 7d) = 164$ (1); simplify to linear equations in a, d (1); $a = 3$, $d = 5$, AP $3, 8, 13, 18, \ldots$ (1).	1+1+1
18	$d = 7$, $a = 4$ (2); $S_{18} = 1143$ (1).	2+1
	OR: Terms form an AP with $d = 11$ (1); count and endpoints (1); sum $= 7062$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 20, 22, 24 (1). (ii) $a_{15} = 20 + 14 \times 2 = 48$ (1). (iii) $S_{15} = \frac{15}{2}(20 + 48) = 510$ (1). (iv) $a_8 = 20 + 7 \times 2 = 34$ seats (1); collection $= 510 \times 150 = ₹\,76{,}500$ (1).	1+1+1+2
20	AP with $a = 120$, $l = 48$, $n = 19$ (1); $S_n = \frac{19}{2}(120 + 48) = 1596$ cm (2); middle term $= \frac{120+48}{2} = 84$ cm (2).	1+2+2
	OR: $S_n = \frac{10}{2}[2a - 270] = 1150$ (2); $a = 250$ (2); last prize ₹-20 (1).	2+2+1

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 63	$a_n = S_n - S_{n-1}$
2	**(a)** 10	$d = 2p$
3	**(c)** 1344	Find n first
4	**(b)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
5	**(c)** 0	Common difference is -1; $a_{p+q} = 0$
6	**(c)** 44	$a_n = \frac{2S_n}{n} - a$
7	**(c)** 56	$a_n = S_n - S_{n-1}$
8	**(a)** 2	$d = 2p$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$a_{n+1} - a_n = (4n+3) - (4n-1) = 4$, a constant, which is why it is an AP.
10	**(c)** A is true but R is false.	$S_{15} = \frac{15}{2}(2 \times 1 + 14 \times 2) = 225 = 15^2$, so A is true; the correct formula has $2a$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2a + 8d = 32$, $2a + 9d = 34$ (1); $d = 2$, $a = 8$ (1).	1+1
12	$a + 2d = 14$, $a + 13d = 3$ (1); $d = -1$, $a = 16$ so $a_{17} = 0$ (1).	1+1
13	$a_1 = S_1 = 8$ (1); $S_2 = 26$, $a_2 = 18$, $d = 10$ (1).	1+1
14	$2a + 8d = 50$, $2a + 9d = 56$ (1); $d = 6$, $a = 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{6}{2}(2a + 5d) = 105$ and $\frac{11}{2}(2a + 10d) = 275$ (1); simplify to linear equations in a, d (1); $a = 10$, $d = 3$, AP $10, 13, 16, 19, \ldots$ (1).	1+1+1
16	$d = 2$, $a = 15$ (2); $S_{22} = 792$ (1).	2+1
17	Terms form an AP with $d = 8$ (1); count and endpoints (1); sum $= 870$ (1).	1+1+1
	OR: $\frac{5}{2}(2a + 4d) = 75$ and $\frac{10}{2}(2a + 9d) = 325$ (1); simplify to linear equations in a, d (1); $a = 1$, $d = 7$, AP $1, 8, 15, 22, \ldots$ (1).	1+1+1
18	$d = 6$, $a = 7$ (2); $S_{12} = 480$ (1).	2+1
	OR: Terms form an AP with $d = 13$ (1); count and endpoints (1); sum $= 9265$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 12 and 14 pages (1). (ii) $a_{10} = 10 + 9 \times 2 = 28$ (1). (iii) $S_5 = \frac{5}{2}(20 + 8) = 70$ (1). (iv) $\frac{n}{2}[20 + 2(n-1)] = 360 \Rightarrow n^2 + 9n - 360 = 0 \Rightarrow (n + 24)(n - 15) = 0$, so $n = 15$ days (1); last day $= 10 + 14 \times 2 = 38$ pages (1).	1+1+1+2
20	AP: $a = 800$, $d = 60$, $n = 30$ (1); last $= a + (n-1)d = 2540$ (2); total $= S_n = \frac{30}{2}(800 + 2540) = 50100$ (2).	1+2+2
	OR: $S_n = \frac{12}{2}[2a - 165] = 1410$ (2); $a = 200$ (2); last prize ₹35 (1).	2+2+1

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
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